ART FIBER SYSTEMS CHALLENGE · SCHUBKI NATIONAL AI HACKATHON 2026

AFS Spectroscopy Assistant

A customer enquiry goes in. A cited first technical assessment comes out.

Team Helios

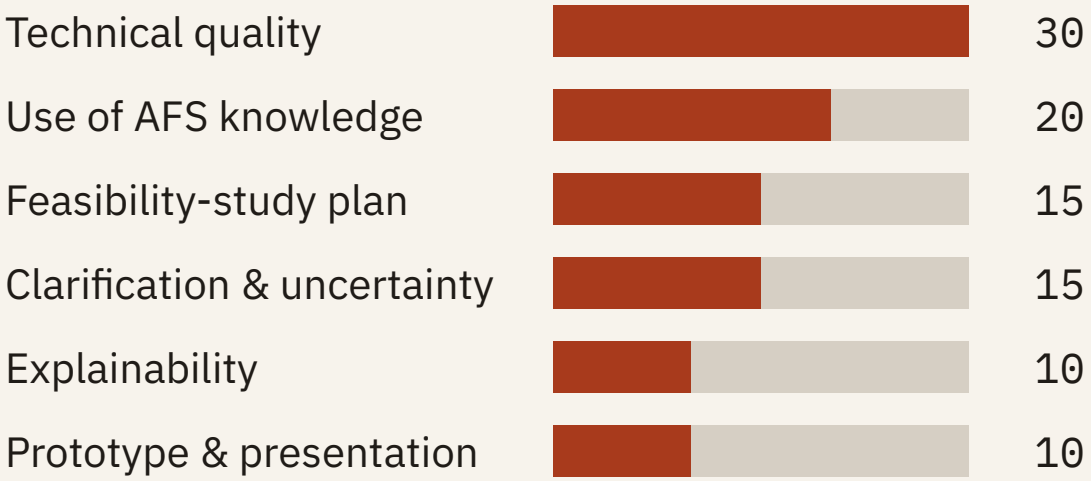
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REPO_URL_PENDING

Help an AFS expert answer a new customer

- A customer describes what they want to measure in their process. **Which method and probe should they test first?**
- The answer must rest on the **AFS documents**. Nothing invented: no prices, accuracy figures or fiber lengths.
- It should **ask the right questions** and **propose a first feasibility test**.

HOW IT IS JUDGED (WEIGHT %)



OUR APPROACH Models write. Code checks every cited claim against the AFS documents.

SYSTEM ARCHITECTURE

Models write. Code checks.

MODEL
Understand the enquiry
Picks out what the customer actually wrote, and what is missing.

→

CODE
Screen the 5 methods
UV-Vis, NIR, mid-IR, Raman, fluorescence, against 44 AFS rules and fiber limits.

→

MODEL
Write the assessment
Reads all 67 AFS pages. Every sourced claim must quote its page.

→

CODE
Check every claim
122 to 161 checks per answer: quotes, pages, numbers, wording.

→

SECOND MODEL
Fix what failed
Gets the list of failures and rewrites. At most 2 rounds.

→

CODE
Answer with sources
Anything still failing is removed or labelled.

☐ language model ■ code: same input, same result

READER
Gemini 3.5 Flash-Lite
Understands the enquiry

WRITER
GLM-5.3-Flash
Writes the first draft

FIXER
Gemini 3.8 Flash
Rewrites, only when checks fail

BACKUP
A different model
Retries if a reply breaks

> **Why this mix:** on the practice cases: quality close to the best (4.06 vs 4.15), **3–6× faster**, about **$0.03** per case.

> Literature only fills gaps the AFS documents leave open, and is marked **secondary**, not AFS material.

How we use the AFS materials

THE AFS DOCUMENTS

Lab report, slide deck, flyer
PDFs (S01, S02, S05)

Two research papers
kidney tissue study (S03), mid-IR + fluorescence (S04)

Guidance document
answer structure and method overview (REF)

Table 1 spreadsheet
methods compared, with a cost rating

A link to the laboratory website (S06)
one line: the web address

WHAT THE CODE BUILDS

67 pages of text
read by the model; every quote is checked against them

44 rules
each tied to an AFS quote, e.g. silica fibers stop near 2.4 µm (S02 p.6)

19 Table 1 rows
cost class: low, average or high (S01 p.6)

7 fiber families
the wavelength range each one carries (S02 p.6)

Same files in, same files out. A check confirms that all 77 quotes behind the rules and fiber ranges are word for word on their page.

FOUR CHECKS ON EVERY CITED CLAIM

1. The quote is on the page, word for word.
2. The page is the right kind of source.
   A study of a similar application counts as precedent, never as proof for this customer.
3. Numbers come from the quote or the customer.
4. No prices, no guarantees, no approval wording.

WHAT THE READER SEES NEXT TO EACH CLAIM

`AFS sources`
the quoted AFS text is on that page, word for word, and the page may support this kind of claim.

`precedent only`
a similar published use: a reason to test, not proof.

`engineering judgement, no source`
our reasoning, clearly marked as such.

120/120
quotes found word for word on the cited page (six evaluation answers)

EXAMPLE RECOMMENDATION · GENERATED BY THE ASSISTANT, NOT EDITED

Reuse of a silica link for petroleum MIR ATR

THE CUSTOMER ASKS (OUR SUMMARY)

The customer wants to measure water in a petroleum line using MIR ATR with a remote analyzer in another building, reusing an existing silica fiber, and asks for concept approval.

Premise not supported: the sources do not support the customer's premise as stated; the assessment below is provisional.

Proposed now: the recommended option (basis: supplied AFS sources): MIR, ATR probe as primary candidate [S02 p.13; S01 p.9].

OPTION	METHODS	COST CLASS, S01 TABLE 1 (P.6)	CONFIDENCE
Simple	NIR, transmission/transflection probe	average	Low
Recommended (proposed)	**MIR, ATR probe**	**high**	**Low**
Extended (multi-method)	NIR, transmission/transflection probe; MIR, ATR probe + NIR (combination)	high	Low

Confidence rates the basis of the advice, not a predicted accuracy; cost class is relative, not a price.

CLARIFICATION QUESTIONS

1. What is the actual link length and routing between the petroleum line and the analyzer building?
2. What is the physical state and condition of the petroleum at the measurement point (liquid, viscous, opaque, temperature, fouling)?
3. What water-content range, accuracy and response time does the investigation require?

The first test plan for the same customer

> **Next step:** Supply the link length, routing and operating conditions, and run the laboratory MIR ATR feasibility test on representative petroleum samples with Karl Fischer reference values.

FIRST FEASIBILITY EXPERIMENT

1. Collect representative petroleum samples spanning the expected water-content range and physical variation (temperature, opacity, phase). (no verified source: engineering judgement)
2. Measure each sample with a laboratory MIR ATR setup (ZrO2 or diamond ATR tip) using a CIR/PIR fiber probe. [18: S02 p.13]
3. Obtain independent reference water-content values for every sample by a laboratory method (e.g. Karl Fischer titration). (no verified source: engineering judgement)
4. Build and validate a chemometric calibration on held-out samples; if MIR ATR alone falls short of the target, test the MIR+NIR combination against the best single method. [19: S01 p.5]
5. Confirm the achievable MIR fiber link length with AFS before fixing the analyzer location; define go/no-go criteria (accuracy, detection limit, response time) with the customer. (no verified source: engineering judgement)

REFERENCE MEASUREMENTS

Karl Fischer titration or an equivalent laboratory water-in-oil method, supplied by the customer.

DECISION BASIS

Go/no-go criteria (required accuracy, detection limit, response time) to be defined numerically by the customer; the plan proceeds only if the MIR link length is confirmed feasible.

RESULTS

The six evaluation cases, answered by the agent

	THE CUSTOMER ASKS	WHAT WAS TRICKY	THE AGENT'S VERDICT
H01	New powder blend: can we reuse our NIR calibration if we add Raman?	A calibration only fits the mixture it was built on.	PREMISE NOT SUPPORTED Rebuild the NIR calibration on the new blend; test Raman separately.
H02	Clear whey and opaque milk concentrate: what should we test first?	One clear stream, one opaque stream.	PROVISIONAL NIR: transmission/transflection for the whey, diffuse reflection for the concentrate.
H03	Can our fluorescence spectrometer report glucose and viable biomass?	Glucose is not fluorescent; no marker named.	PREMISE NOT SUPPORTED Test Raman first, NIR as the comparison.
H04	Reuse a silica cable for MIR ATR on a petroleum line?	Silica fibers cannot carry mid-IR light.	PREMISE NOT SUPPORTED Test MIR ATR, with a CIR or PIR fiber.
H05	A liquid goes off-spec. Give us the exact sensor now.	Contaminant and quality variable unknown.	INSUFFICIENT INFORMATION Lab trial first: Raman and mid-IR ATR on normal and off-spec samples.
H06	Does the kidney study prove our sensor is ready for live surgery?	It is an ex vivo proof of concept; no own data.	PREMISE NOT SUPPORTED Ex vivo NIR test on own tissue, with histopathology as reference.

Run once with the final version; answers not edited. A separate review agent checked all six against the AFS sources: each reaches the right core verdict.

6/6
answered

120/120
quotes verified on the cited page

0
failed hard checks left in any answer (8 minor warnings across the six)

1.5–3 min
per case · $0.29 for all six

LIMITATIONS

Limitations and next steps

TODAY		NEXT
Full answers are about 3,000 words; our short view (slide 5) is 560–698 words; AFS's own worked answers are 300–400.	→	A customer reply as short as AFS writes them
A quote counts as verified when it is on the cited page. That does not yet prove it supports the sentence.	→	A second checker: does the quote really support the sentence?
15 of the 17 options in the six evaluation answers say "Low".	→	Confidence that moves with what is known
One of our own complete test enquiries still got 4–5 questions.	→	Fewer questions when the enquiry is already complete
The fixing step takes 46–72 % of each run.	→	Faster runs

Demo video (3:27): docs/video/demo.mp4 in the repository